Some Properties of Prime-Cubed Cyclotomic Cosets in Cyclic Groups

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A **group** is a set equipped with a binary operation that combines any two elements to form a third element in such a way that four conditions called group axioms are satisfied, namely closure, associativity, identity and invertibility.

A **cyclic group**, a group that is generated by a single element, is an interesting structure in algebra.

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A set of integers modulo m is denoted by $\mathbb{Z}_m$.

The set $\mathbb{Z}_m$ together with the addition modulo m is known to be a cyclic group.

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A **prime power** q is an integer that can be written in the form of $q = p^k$, where p is a prime and k is a positive integer.

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For each $a \in \mathbb{Z}_m$ and prime power q such that $\gcd(q, m) = 1$, the q**-cyclotomic coset of** $\mathbb{Z}_m$ **containing** a is defined to be

$$S_q(a) := \{q^i \cdot a \mid i = 0, 1, 2, \dots\},$$

where $q^i \cdot a := \sum_{j=1}^{q^i} a$ in $\mathbb{Z}_m$.

For convenience, denote by $q^i a$ the element $q^i \cdot a \in \mathbb{Z}_m$.

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Types of q-cyclotomic cosets where $q = 2^m$ have been long considered due to their wide application in coding theory. Studying q-cyclotomic cosets of different types where $q = p^m$ for any prime p and a positive integer m gives us tools to characterize and enumerate self-orthogonal and complementary dual codes.

In this research, we consider $m = 3$ and introduce six types of q-cyclotomic cosets. Precisely, properties, characterizations and enumerations of such q-cyclotomic cosets will be presented in the case where $q = p^3$ is a **prime-cubed**.

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Let us recall the definition of q-cyclotomic coset of $\mathbb{Z}_m$ containing a, $S_q(a)$:

For each $a \in \mathbb{Z}_m$ and prime power q such that $\gcd(q, m) = 1$,

$$S_q(a) = \{q^i a \mid i = 0, 1, 2, \dots\}.$$

Example

The 8-cyclotomic coset of $\mathbb{Z}_{45}$ containing 0 is $S_8(0) = \{8^i \cdot 0 \mid i = 0, 1, 2, \dots\} = \{0\}$ and the 8-cyclotomic coset of $\mathbb{Z}_{45}$ containing 1 is

$$S_8(1) = \{8^i \cdot 1 \mid i = 0, 1, 2, \dots\} = \{1, 8, 19, 17, 1, 8, \dots\} = \{1, 8, 17, 19\}.$$

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$= \{1, 8, 17, 19\}$.

From $S_8(a) = \{8^i a \mid i = 0, 1, \dots\}$. All 8-cyclotomic cosets in $\mathbb{Z}_{45}$ are $S_8(0) = \{0\}$,

$S_8(1) = S_8(8) = S_8(17) = S_8(19) = \{1, 8, 17, 19\}$,
$S_8(2) = S_8(16) = S_8(34) = S_8(38) = \{2, 16, 34, 38\}$,
$S_8(3) = S_8(6) = S_8(12) = S_8(24) = \{3, 6, 12, 24\}$,
$S_8(4) = S_8(23) = S_8(31) = S_8(32) = \{4, 23, 31, 32\}$,
$S_8(5) = S_8(40) = \{5, 40\}$,
$S_8(7) = S_8(11) = S_8(29) = S_8(43) = \{7, 11, 29, 43\}$,
$S_8(9) = S_8(18) = S_8(27) = S_8(36) = \{9, 18, 27, 36\}$,
$S_8(10) = S_8(35) = \{10, 35\}$,
$S_8(13) = S_8(14) = S_8(22) = S_8(41) = \{13, 14, 22, 41\}$,
$S_8(15) = S_8(30) = \{15, 30\}$,
$S_8(20) = S_8(25) = \{20, 25\}$,
$S_8(21) = S_8(33) = S_8(39) = S_8(42) = \{21, 33, 39, 42\}$,
$S_8(26) = S_8(28) = S_8(37) = S_8(44) = \{26, 28, 37, 44\}$.

Proposition ([1])

The q-cyclotomic cosets of $\mathbb{Z}_m$ form a partition of $\mathbb{Z}_m$.

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The q-cyclotomic cosets of $\mathbb{Z}_m$ form a partition of $\mathbb{Z}_m$.

Definition

Let $a \in \mathbb{Z}_m$ and $q = p^3$ be a prime-cubed number. A q-cyclotomic coset $S_q(a)$ is said to be of

type I	if	$S_q(a) = S_q(-a)$	or
type I′	if	$S_q(a) \neq S_q(-a)$,	
type II	if	$S_q(a) = S_q(-pa)$	or
type II′	if	$S_q(a) \neq S_q(-pa)$	

and

type III	if	$S_q(a) = S_q(-p^2a)$	or
type III′	if	$S_q(a) \neq S_q(-p^2a)$.	

Note that if $S_q(a)$ is of type I then it is not of type I′ and vice versa, as well as type II and III.

Example For the 8-cyclotomic cosets of $\mathbb{Z}_{45}$:

$$\begin{aligned} S_8(9) &= \{9,18,27,36\}, & S_8(-9) &= \{9,18,27,36\} \\ & & S_8(-2\cdot 9) &= \{9,18,27,36\} \\ & & S_8(-2^2\cdot 9) &= \{9,18,27,36\}. \end{aligned}$$

$$\begin{aligned} S_8(5) &= \{5,40\}, & S_8(-5) &= \{5,40\} \\ & & S_8(-2\cdot 5) &= \{10,35\} \\ & & S_8(-2^2\cdot 5) &= \{20,25\}. \end{aligned}$$

$$\begin{aligned} S_8(1) &= \{1,8,17,19\}, & S_8(-1) &= \{26,28,37,44\} \\ & & S_8(-2\cdot 1) &= \{7,11,29,43\} \\ & & S_8(-2^2\cdot 1) &= \{13,14,22,41\}. \end{aligned}$$

So $S_8(9)$ is of types I, II and III,
$S_8(5)$ is of types I, II′ and III′ and
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So $S_8(9)$ is of types I, II and III,
$S_8(5)$ is of types I, II$'$ and III$'$ and
$S_8(1)$ is of types I$'$, II$'$ and III$'$.

Because each $S_q(a)$ in $\mathbb{Z}_m$ has different properties depended on variables q, a and m, the study gives different results. Nevertheless, our main questions, for given q and m, are:

Q1. How are the types of $S_q(a)$ relate to each other?
Q2. How can we characterize the types of $S_q(a)$ for a given a?
Q3. How can we enumerate $S_q(a)$ of each type?

Some parts of questions has been answered in references:

A2. The characterization of $S_q(a)$ of type I has been given by a characteristic function.
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So we study the rest of questions in this research which is about types II and III.

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Definition

For $a \in \mathbb{Z}_m$, denote by $\operatorname{ord}(a)$ the **additive order** of a in $\mathbb{Z}_m$ and when $\gcd(a, m) = 1$, denote by $\operatorname{ord}_m(a)$ the **multiplicative order** of a.

Definition

The Euler's totient function ϕ is a function defined by

$$\phi(n) := |\{k \mid 1 \leq k \leq n \text{ and } \gcd(n, k) = 1\}|$$

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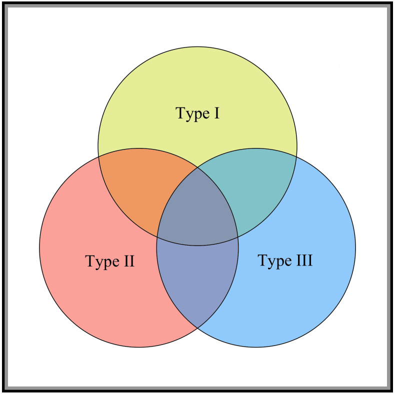
We introduce a simple Venn diagram of types.



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Q1. How are the types of $S_q(a)$ relate to each other?

Theorem 1

Let q be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. For each $a \in \mathbb{Z}_m$,

$$S_q(a) \text{ is of type II if and only if } S_q(a) \text{ is of type III.}$$

Theorem 2

Let q be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. For each $a \in \mathbb{Z}_m$.

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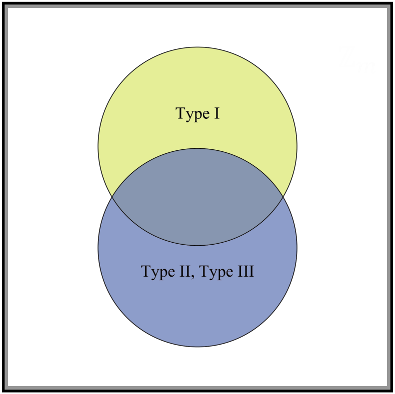

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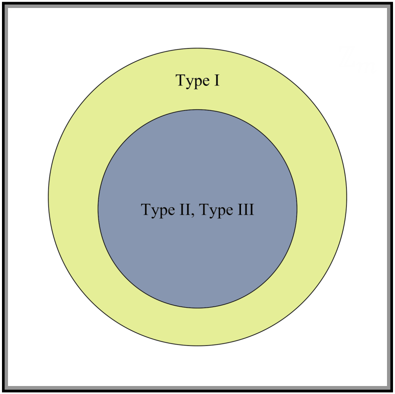

Question

Q2. How can we characterize the types of $S_q(a)$ for a given a?

A2. The characterization of $S_q(a)$ of type I has been given by a characteristic function.

Lemma

Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$, then the q-cyclotomic coset

$$S_q(a) \text{ is of type II if and only if } \operatorname{ord}(a) \mid p^{3s+1} + 1$$

for some integer $s \geq 0$.

$S_q(a)$ is of type II if and only if $\mathrm{ord}(a) \mid p^{3s+1}+1$

Definition

For each prime p, let $\lambda_p : \mathbb{N} \to \{0,1\}$ be a function defined by

$$\lambda_p(t) = \begin{cases} 1 & \text{if there exists } s \geq 0 \text{ such that } t \mid p^{3s+1}+1, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 3

Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m,q)=1$. Let $a \in \mathbb{Z}_m$. Then the q-cyclotomic coset

$$S_q(a) \text{ is of type II (and also III) if and only if } \lambda_p(\mathrm{ord}(a)) = 1.$$

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Question

Q3. How can we enumerate $S_q(a)$ of each type?

A3. The enumeration of type I as well as I$'$ had been studied.

Theorem 4

Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Then the number of q-cyclotomic cosets of type II and type III is

$$\sum_{d|m} \lambda_p(d) \frac{\phi(d)}{\mathrm{ord}_d(q)},$$

and the number of type II$'$ and III$'$ is

$$\sum_{d|m} (1 - \lambda_p(d)) \frac{\phi(d)}{\mathrm{ord}_d(q)}.$$

Corollary 5

Let a be a positive integer and p be a prime number. The following statements are equivalent:

(1) $a \mid p^{3s-2} + 1$ for some integer $s \geq 0$.
(2) $a \mid p^{3s-1} + 1$ for some integer $s \geq 0$.
(3) $a \mid p^{3s+1} + 1$ for some integer $s \geq 0$.
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This number theoretical result can be proved by just only the properties of prime-cubed cyclotomic cosets.

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Thank you for your attention.